EE115B *Digital Circuits*

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- Prof. Zhifeng Zhu
- Prof. Yajun Ha

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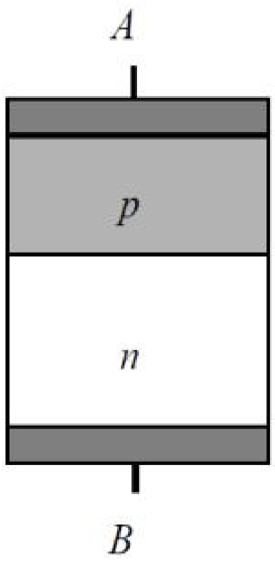
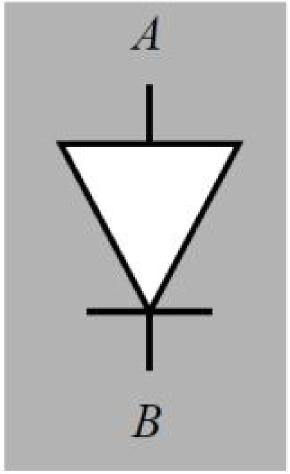
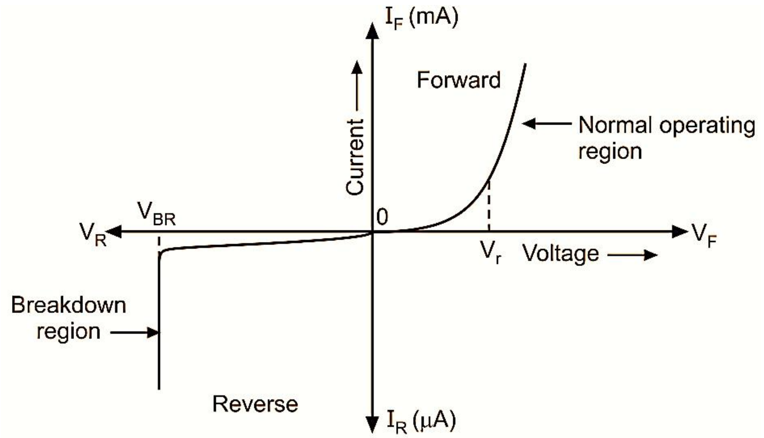
Review



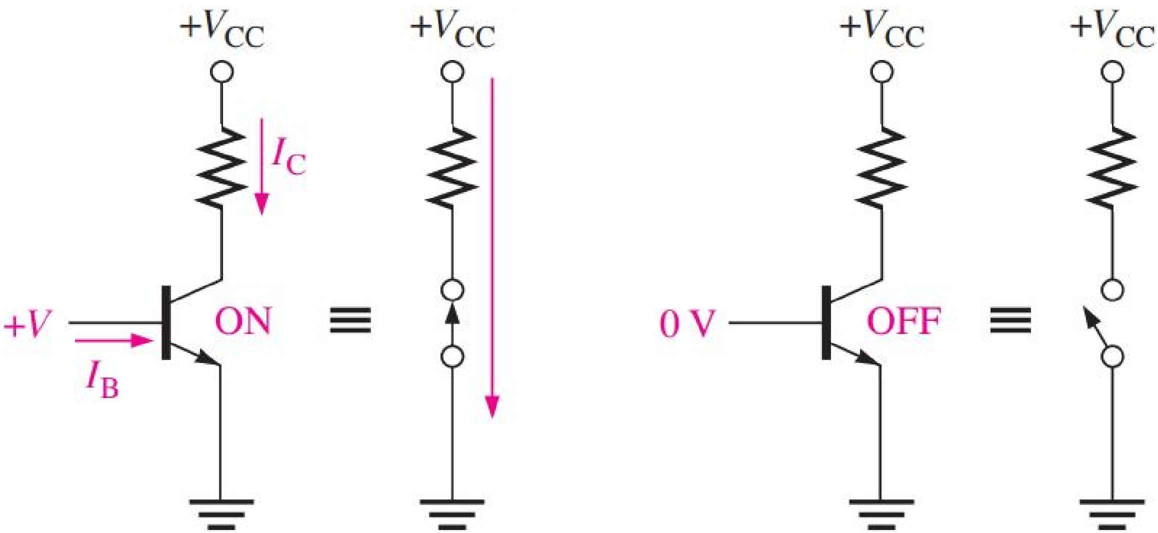
Schematics



Circuit symbol

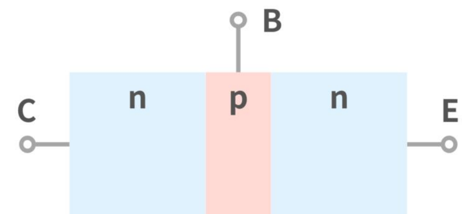


https://howelectrical.com/vi-characteristics-of-pn-junction-diode/



(a) Saturated (ON) transistor and ideal switch equivalent

(b) OFF transistor and ideal switch equivalent



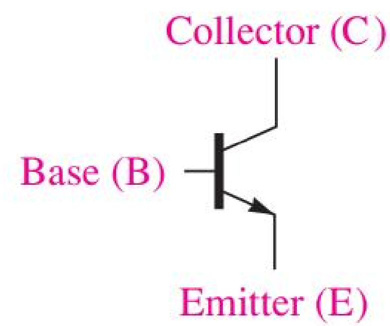


FIGURE 15–25 The symbol for a BJT.

VBE>0.7 ON
VBE<0.7 OFF

Review

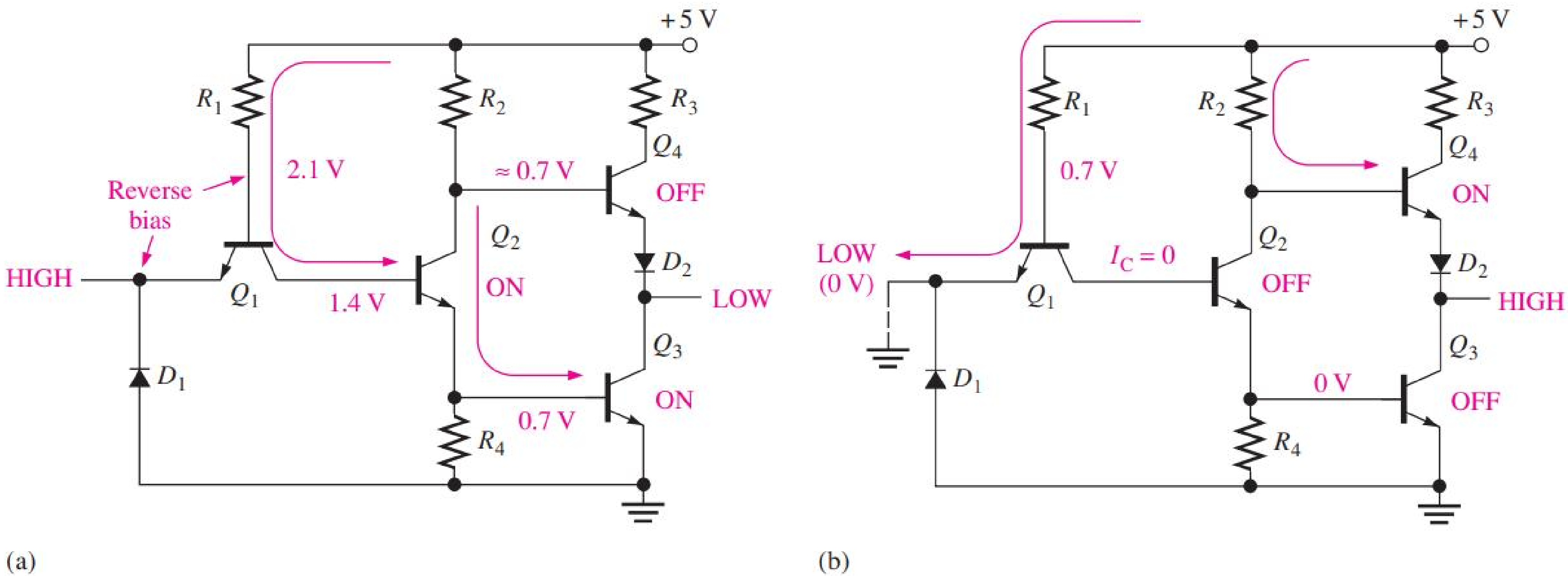
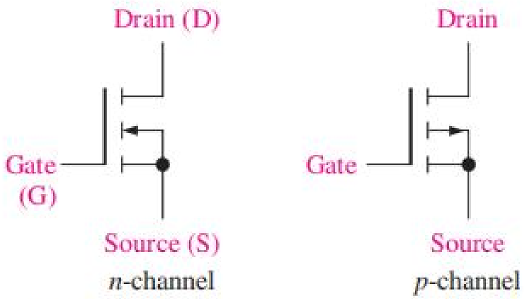
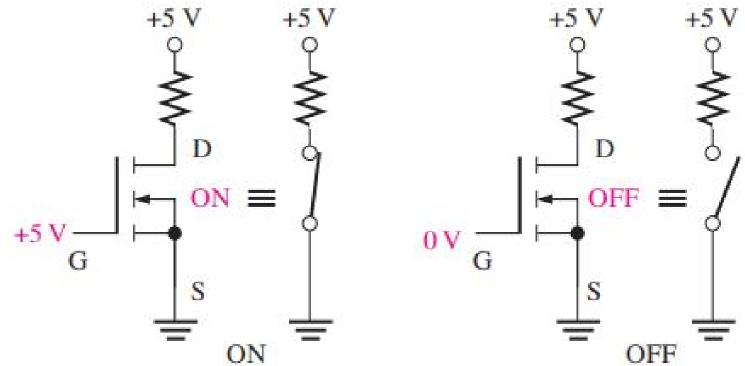


FIGURE 15–28 Operation of a TTL inverter.

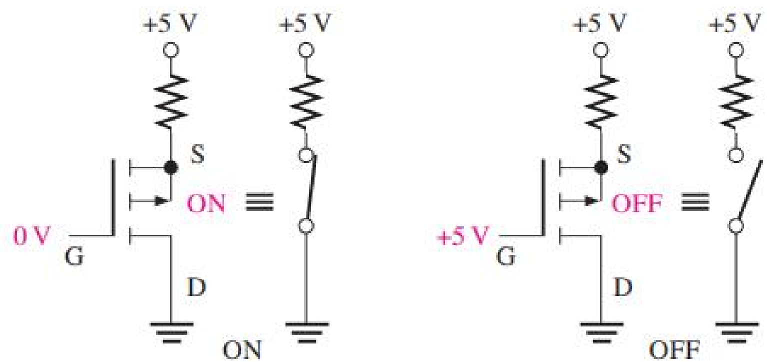
Review



(a) MOSFET symbols



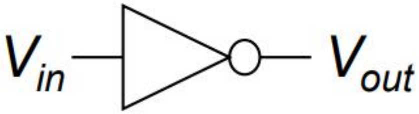
(b) *n*-channel switch



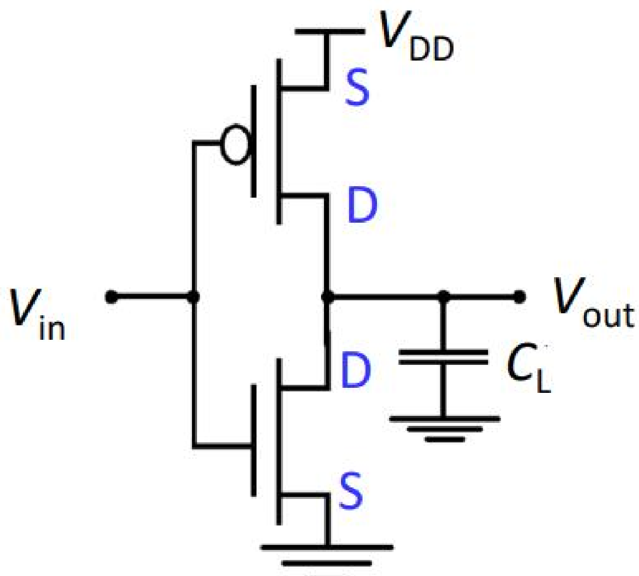
(c) *p*-channel switch

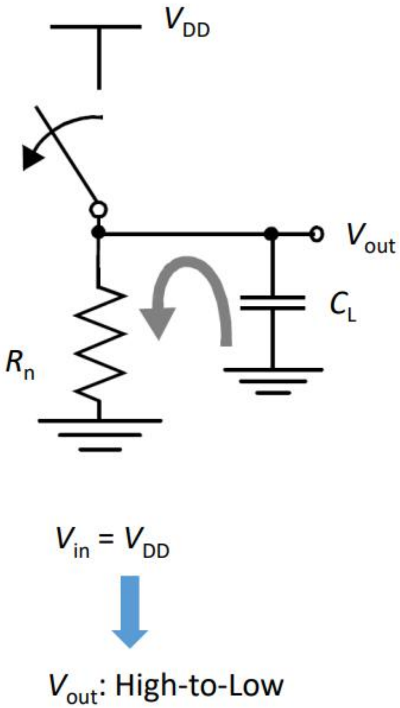
Review

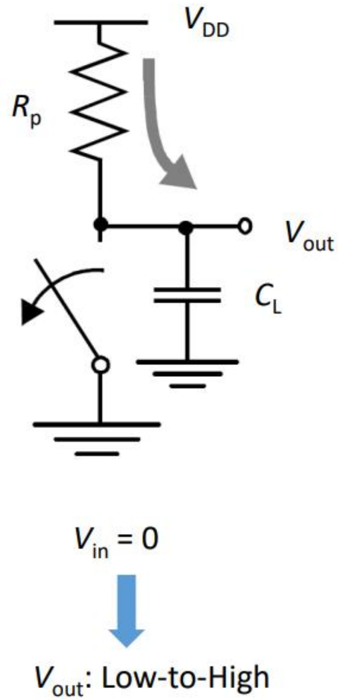
Symbol



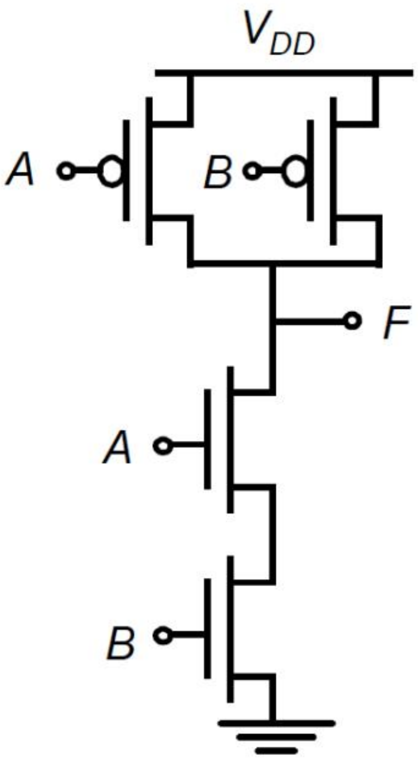
Schematic





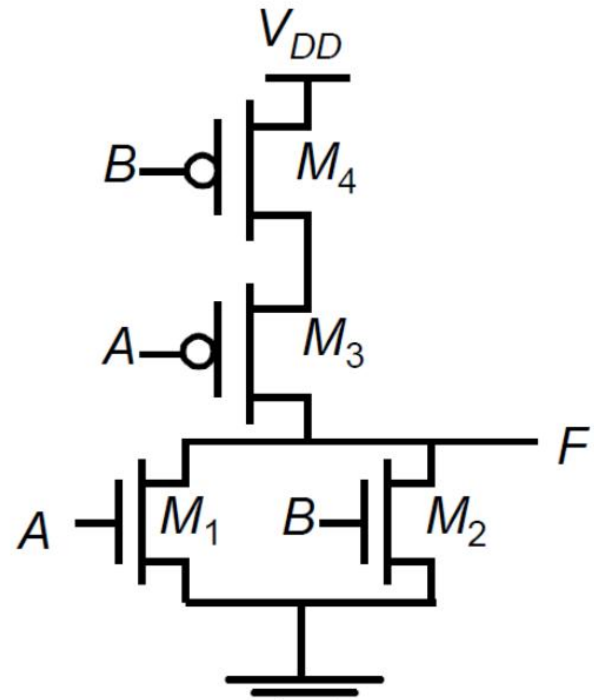


Review



Circuit

NAND

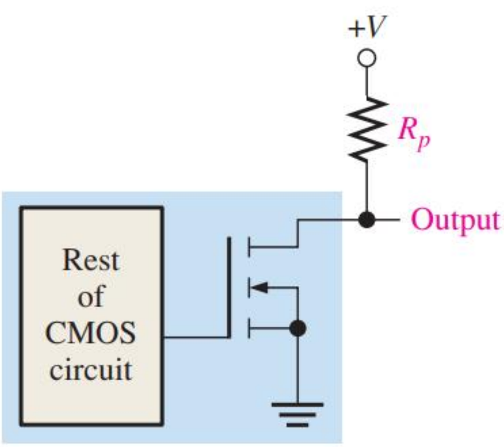


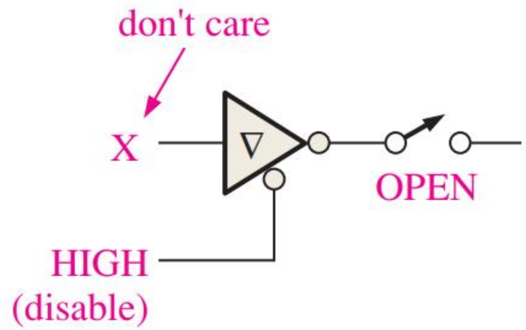
Circuit

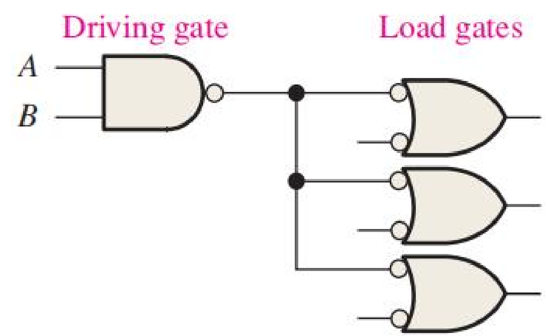
NOR

Review

- Open Collector/Open Drain
- Tristate gates
- Fan-out
- Source current/Sink Current







Boolean Algebra and Logic Simplification

- Boolean Algebra
- Logic Simplification
  - Karnaugh Map
  - Quine-McCluskey Method

What is SOP Form/standard SOP form

- All Boolean expressions can be converted into either sum-of-product (SOP) form or product-of-sum (POS) form.
- In SOP, a single overbar cannot extend over more than one variable.
- Standard SOP is one in which all the variables appear in each term

Question: Are they standard SOP expression?

$AB+ABC$

$ABC+CDE+B'CD'$

$A'B+A'BC'+AC$

$AB'CD+A'B'CD'+ABC'D'$

$AB+A'B+AB'+A'B'$

Convert to Standard SOP

1. Multiply each nonstandard product term by the sum of a missing variable and its complement since *A* + *A'* = 1.
2. Repeat step 1 until all terms are in standard form

$$A\overline{B}C + \overline{A}\,\overline{B} + AB\overline{C}D$$

Question: Convert *A*+*B'C* to standard SOP

① $A\overline{B}C = A\overline{B}C(D + \overline{D}) = A\overline{B}CD + A\overline{B}C\overline{D}$

② $\overline{A}\,\overline{B} = \overline{A}\,\overline{B}(C + \overline{C}) = \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}\,\overline{C}$

③ $\overline{A}\,\overline{B} = \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}\,\overline{C} = \overline{A}\,\overline{B}C(D + \overline{D}) + \overline{A}\,\overline{B}\,\overline{C}(D + \overline{D})$
$= \overline{A}\,\overline{B}CD + \overline{A}\,\overline{B}C\overline{D} + \overline{A}\,\overline{B}\,\overline{C}D + \overline{A}\,\overline{B}\,\overline{C}\,\overline{D}$

$$A\overline{B}C + \overline{A}\,\overline{B} + AB\overline{C}D = A\overline{B}CD + A\overline{B}C\overline{D} + \overline{A}\,\overline{B}CD + \overline{A}\,\overline{B}C\overline{D} + \overline{A}\,\overline{B}\,\overline{C}D + \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + AB\overline{C}D$$

- Binary representation of a standard product term

$A\overline{B}C\overline{D} = 1 \cdot \overline{0} \cdot 1 \cdot \overline{0} = 1 \cdot 1 \cdot 1 \cdot 1 = 1$ The product term has a binary value of 1010.

Minterm m10
minterm is a product of literals in which each input variable appears exactly once

What is POS Form

A standard POS expression is one in which all the variables in the domain appear in each sum term in the expression.

$$(\overline{A} + \overline{B} + \overline{C} + \overline{D})(A + \overline{B} + C + D)(A + B + \overline{C} + D)$$

Convert to Standard POS

Question: Are they standard POS expression?

$$(\overline{A} + B)(A + \overline{B} + C)$$

$$(\overline{A} + \overline{B} + \overline{C})(C + \overline{D} + E)(\overline{B} + C + D)$$

$$(A + B)(A + \overline{B} + C)(\overline{A} + C)$$

$$(\overline{A} + \overline{B} + \overline{C} + \overline{D})(A + \overline{B} + C + D)(A + B + \overline{C} + D)$$

Convert to Standard POS

1. Add to each nonstandard term the product of the missing variable and its complement since $A \cdot A' = 0$.
2. Repeat Step 1 until all terms are in standard form

$$(A+\overline{B}+C)(\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)$$

Question: Why they are equivalent?

1 $$A+\overline{B}+C = A+\overline{B}+C+D\overline{D} = (A+\overline{B}+C+D)(A+\overline{B}+C+\overline{D})$$

2 $$\overline{B}+C+\overline{D} = \overline{B}+C+\overline{D}+A\overline{A} = (A+\overline{B}+C+\overline{D})(\overline{A}+\overline{B}+C+\overline{D})$$

$$(A+\overline{B}+C)(\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D) = (A+\overline{B}+C+D)(A+\overline{B}+C+\overline{D})(A+\overline{B}+C+\overline{D})(\overline{A}+\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)$$

- Binary representation of a standard sum term

Question: Convert $A(B'+C)$ to standard SOP

$$A+\overline{B}+C+\overline{D} = 0+\overline{1}+0+\overline{1} = 0+0+0+0 = 0$$

The sum term has a binary value of 0101.

For maxterm, complement of variable is counted as 1

12

Maxterm M5

Maxterm is the sum of N distinct literals where each literals occurs exactly once.

Rule 12: $(A+B)(A+C) = A+BC$

Convert SOP to POS

1. Obviously: not(mx)=Mx

2. Demorgan: not(m0+m2+m4)=M0 M2 M4

- Sum-of-products
  - $F' = A'B'C' + A'BC' + AB'C'$
- Apply de Morgan's
  - $(F')' = (A'B'C' + A'BC' + AB'C')'$
  - $F = (A + B + C)\ (A + B' + C)\ (A' + B + C)$

3. Demorgan:not(M1 M3 M5 M6 M7)=m1+m3+m5+m6+m7

- Product-of-sums
  - $F' = (A+B+C')(A+B'+C')(A'+B+C')(A'+B'+C)(A'+B'+C')$
- Apply de Morgan's
  - $(F')' = ((A+B+C')(A+B'+C')(A'+B+C')(A'+B'+C)(A'+B'+C'))'$
  - $F = A'B'C + A'BC + AB'C + ABC' + ABC$

https://courses.cs.washington.edu/courses/cse370
/97au/admin/Slides/Week2Lecture3/tsld008.htm

Convert SOP to POS

Sum of all minterms equals 1:

m0+m2+m3+m5+m7 + (m1+m4+m6)=1

Can you write it immediately?

not(m1+m4+m6) = *M1 M4 M6* = *m0+m2+m3+m5+m7*

$$\overline{A}\,\overline{B}\,\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}C + ABC$$

1. $000 + 010 + 011 + 101 + 111$

2. The remaining terms are 001, 100 and 110, these are the binary values that make the sum term 0.

3. The equivalent POS form is

$$(A + B + \overline{C})(\overline{A} + B + C)(\overline{A} + \overline{B} + C)$$

- Using a similar procedure, you can go from POS to SOP.

SOP, POS, TruthTable

Enumerate all 1 entries

SOP Implementation from a Truth Table

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

$$F = A\overline{B}C + \overline{A}BC + \overline{A}\overline{B}C$$

electroniclinic.com

https://www.electroniclinic.com/sop-and-pos-digital-logic-designing-with-solved-examples/

SOP, POS, TruthTable

Enumerate all 0 entries

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

POS implementation from a Truth Table

$$F = (\overline{A}+\overline{B}+\overline{C})(\overline{A}+\overline{B}+C)(\overline{A}+B+C)(A+\overline{B}+C)(A+B+C)$$

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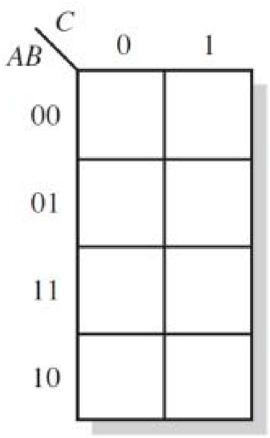
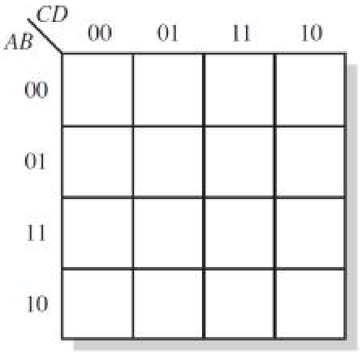
https://www.electroniclinic.com/sop-and-pos-digital-logic-designing-with-solved-examples/

Karnaugh Maps

- The Karnaugh map provides a cookbook method for simplification.
- The idea of Karnaugh map is to use adjacency to simplify the expression
- The Karnaugh map method is limited to 4 variables or below



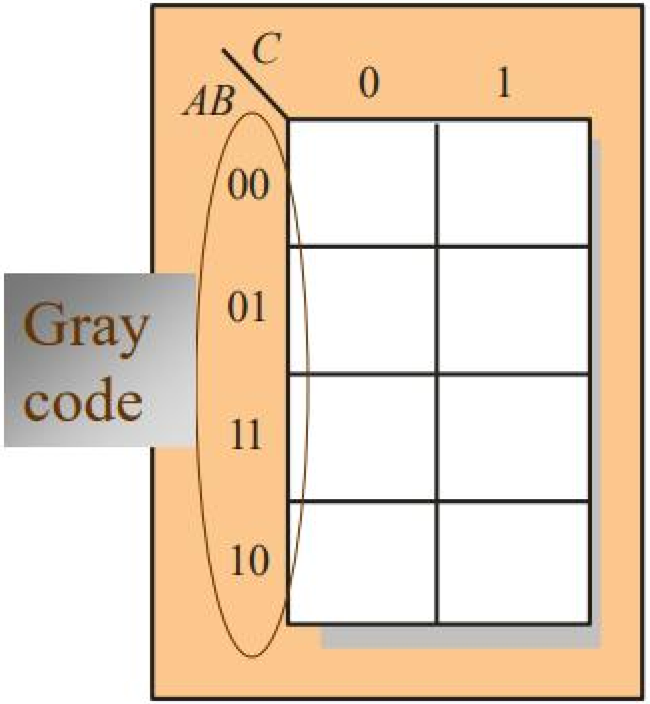
3-variable Karnaugh map



4-variable Karnaugh map

Karnaugh Maps

Cells are usually labeled using 0's and 1's to represent the variable and its complement.



The numbers are entered in gray code, to force adjacent cells to be different by only one variable.

Ones are read as the true variable and zeros are read as the complemented variable.

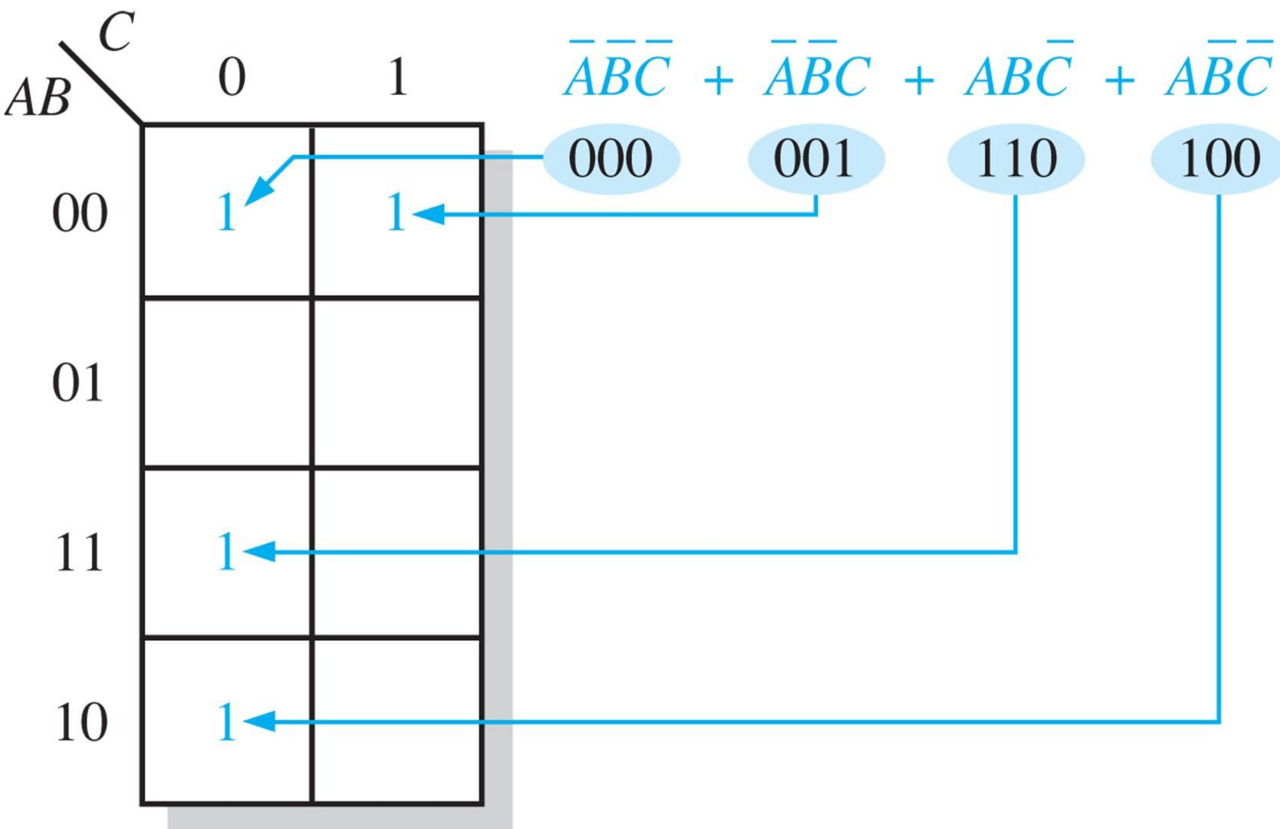
Decimal	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

electronicsarea.com

(3 variables)

Mapping a standard SOP expression

Place a "1" for each minterm.



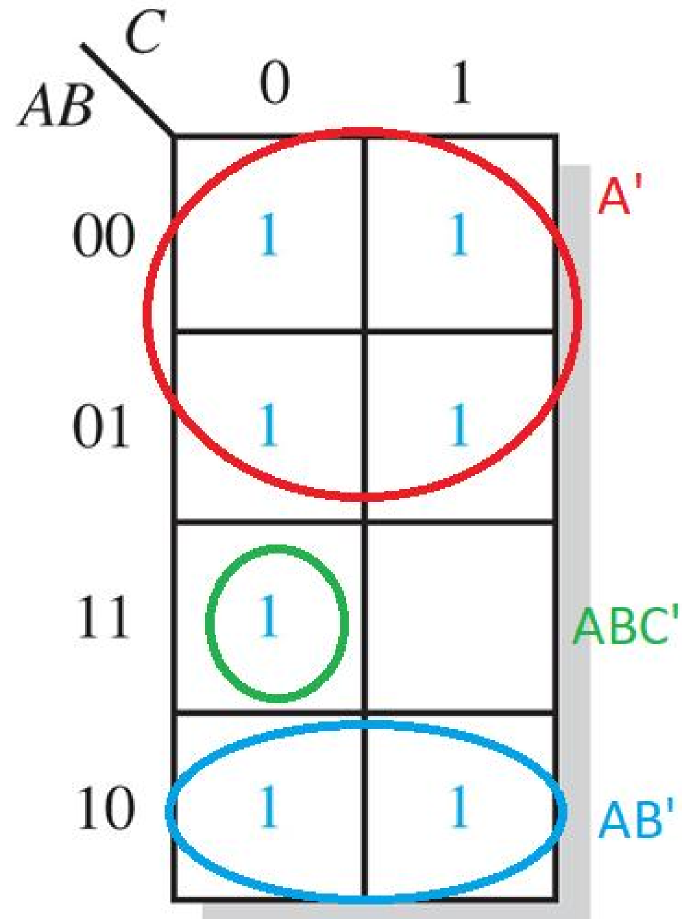
Mapping a nonstandard SOP expression

Expand product terms to minterms and place a “1” for each minterm.

Example: Map $\overline{A} + A\overline{B} + AB\overline{C}$.

Solution:

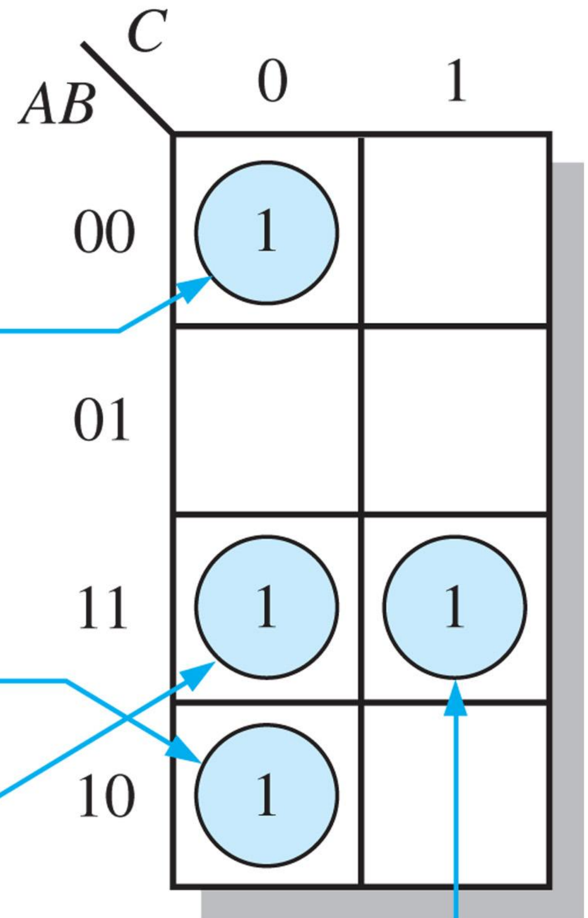
$\overline{A}$	$+ A\overline{B}$	$+ AB\overline{C}$
000	100	110
001	101	
010		
011		



Mapping directly from a truth table

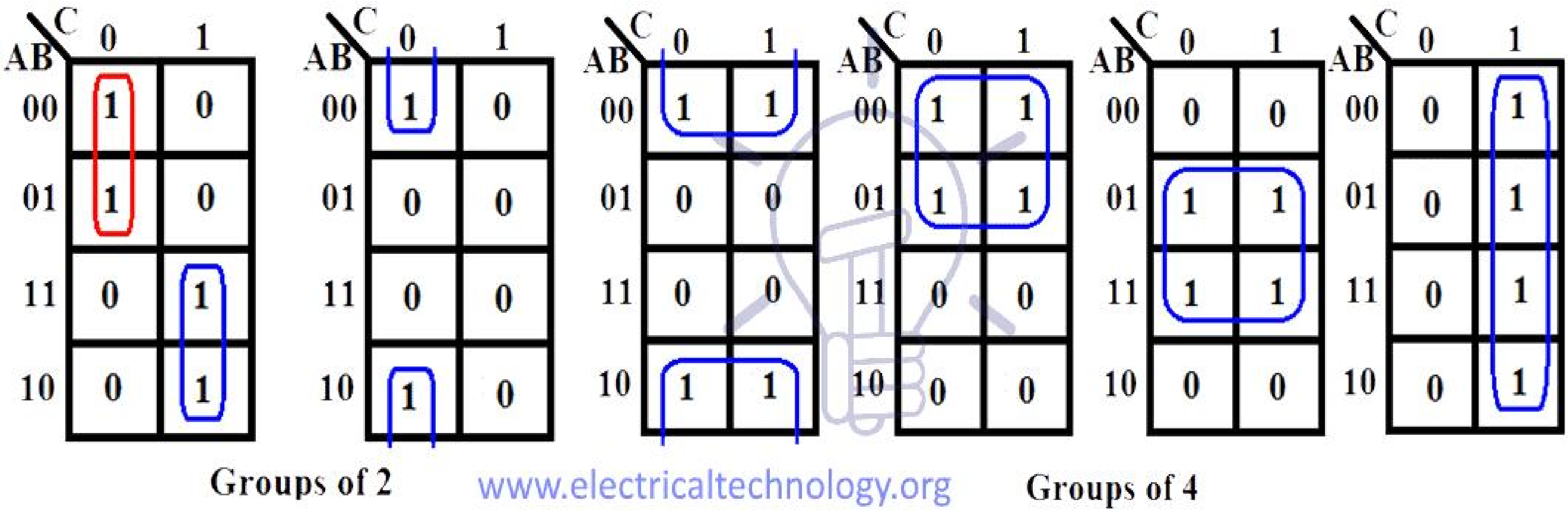
$$X = \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}\,\overline{C} + AB\overline{C} + ABC$$

Inputs			Output
A	*B*	*C*	*X*
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



Simplification as Grouping

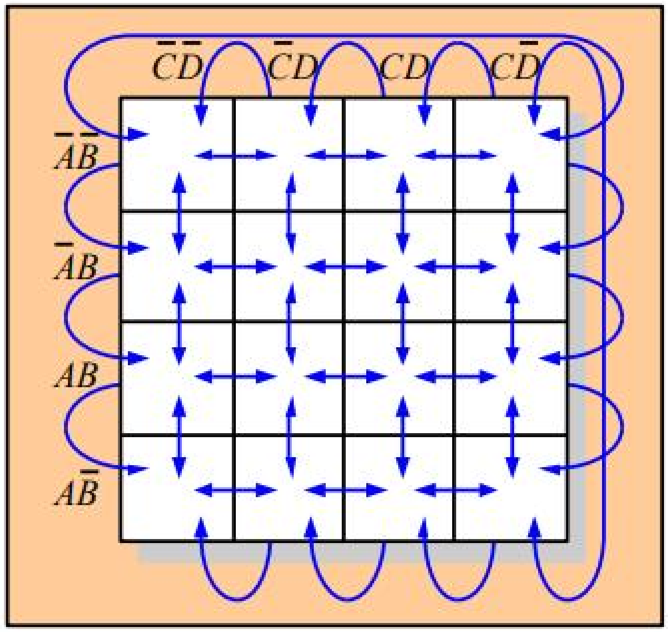
Simplification is achieved by summation of minterms grouped



https://www.electricaltechnology.org/2018/05/karnaugh-map-k-map.html

4 Variable Karnaugh Maps

A 4-variable map has an adjacent cell on each of its four boundaries as shown.



Each cell is different only by one variable from an adjacent cell.

Grouping follows the rules given in the text.

The following slide shows an example of reading a four variable map using binary numbers for the variables…

Simplification as Grouping

BC \ DE	00	01	11	10
00	1	0	0	0
01	0	0	0	1
11	0	0	0	1
10	1	0	0	0

BC \ DE	00	01	11	10
00	1	0	0	0
01	0	0	0	1
11	0	0	0	1
10	1	0	0	0

BC \ DE	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	0	1	0
10	1	0	0	0

BC \ DE	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	0	1	0
10	1	0	0	0

BC \ DE	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

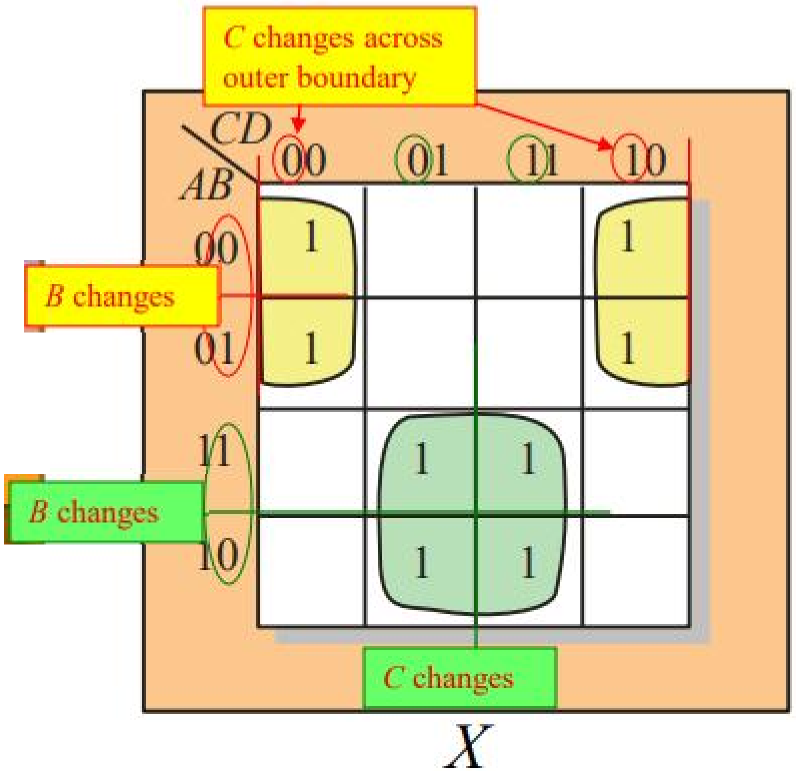
BC \ DE	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

BC \ DE	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

BC \ DE	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

Karnaugh Maps

Example Group the 1's on the map and read the minimum logic.



Solution

1. Group the 1's into two separate groups as indicated.
2. Read each group by eliminating any variable that changes across a boundary.
3. The upper (yellow) group is read as $\overline{A}\overline{D}$.
4. The lower (green) group is read as AD.

$$X = \overline{A}\overline{D} + AD$$

"Don't care" conditions

"Don't care" term can be "1" or "0". Denoted by "X".

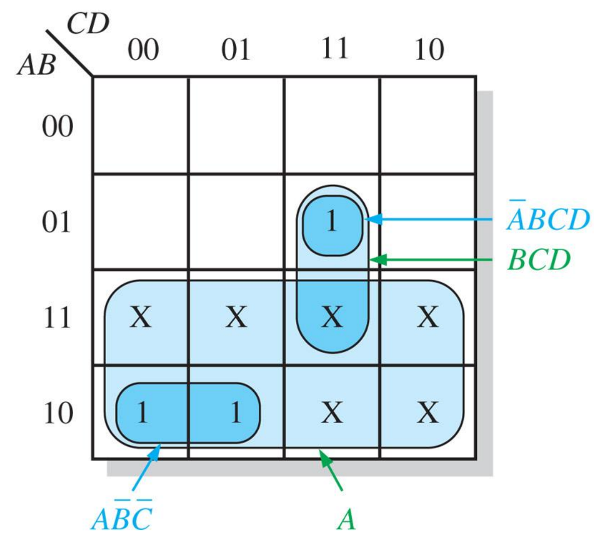
Example: Output is "1" when input BCD code is 7, 8, or 9.

$Y=\sum m(7,8,9) + D(10,11,12,13,14,15)$

Inputs				Output
A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Don't cares

(a) Truth table



(b) Without "don't cares" $Y = A\bar{B}\bar{C} + \bar{A}BCD$
With "don't cares" $Y = A + BCD$

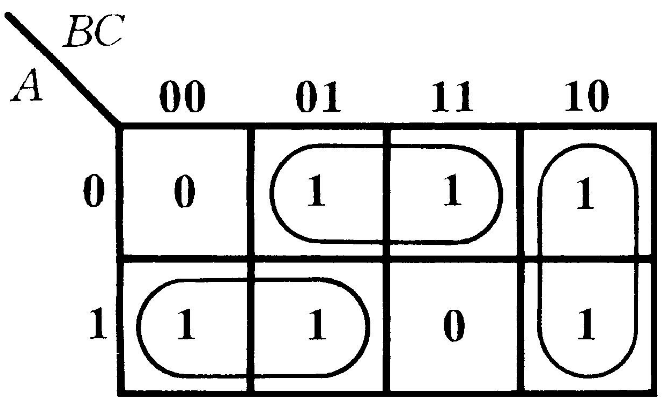
Determine the minimum SOP expression by K-map.

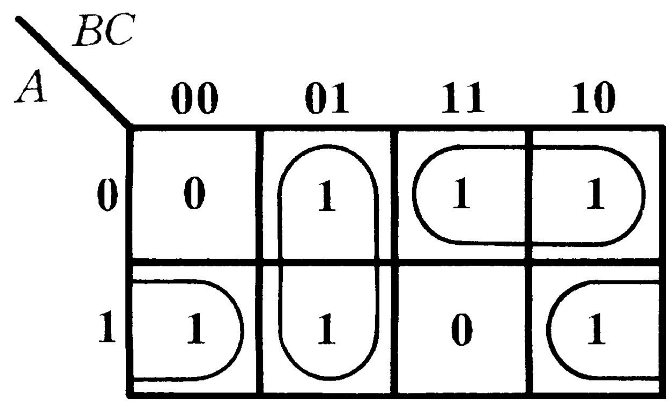
$$Y(A,B,C) = \quad AC' + A'C + B'C + BC'$$

Exercises:simplify SOP

Determine the minimum SOP expression.

$$Y(A,B,C) = AC' + A'C + B'C + BC'$$





$$AB' + A'C + BC'$$

$$AC' + A'B + B'C$$

The final results are not unique. However, the number of terms and the number of variables in each term are identical

Exercises:simplify SOP

Determine the minimum SOP expression by K-map.

$$Y = ABC + ABD + AC'D + C'D' + AB'C + A'CD'$$

Exercises:simplify SOP

Determine the minimum SOP expression.

$$Y = \quad ABC + ABD + AC'D + C'D' + AB'C + A'CD'$$

1. Work on 1's

2. Work on 0's
   (a) Take 0's as 1's (i.e., invert truth table or K-map)
   (b) Write SOP for these 1's to get Y'
   (c) Invert Y' to get Y

Y'=A'D
Y=(Y')'=A+D'

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	1	1	1
10	1	1	1	1

$$A + D'$$

(this is a special case, since Y' needs to be a single product term,otherwise the result Y is not SOP)

Simplification of POS expressions

Use a Karnaugh map to minimize the following standard POS expression:

$$(A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$$

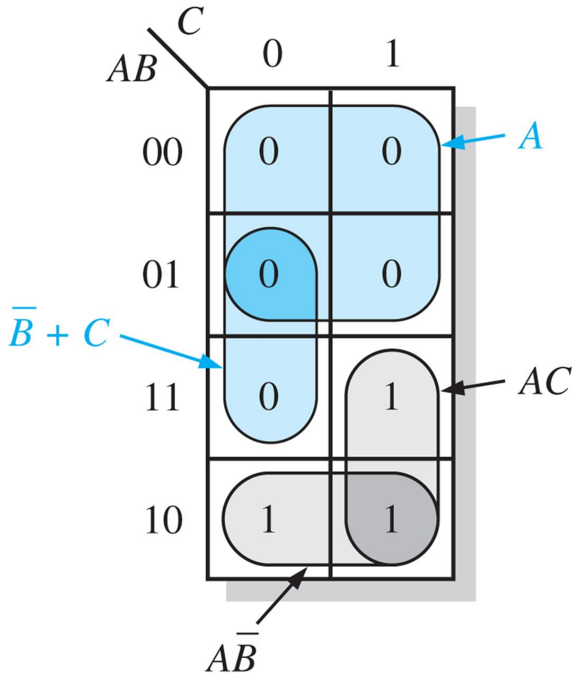
Also, derive the equivalent SOP expression.

Solution: (1) Minimum POS: group 0's

$$A(\overline{B} + C)$$

(2) Minimum SOP: group 1's

$$AC + A\overline{B}$$



Exercise: simplify POS

Use a Karnaugh map to minimize the following POS expression:

$$(B + C + D)(A + B + \overline{C} + D)(\overline{A} + B + C + \overline{D})(A + \overline{B} + C + D)(\overline{A} + \overline{B} + C + D)$$

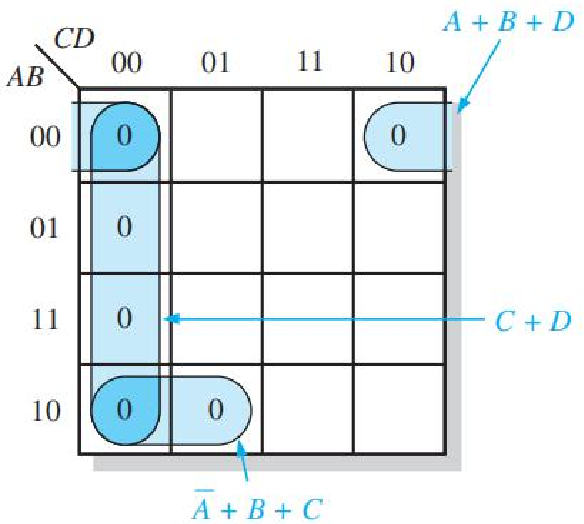
Exercise: simplify POS

$$(B + C + D)(A + B + \overline{C} + D)(\overline{A} + B + C + \overline{D})(A + \overline{B} + C + D)(\overline{A} + \overline{B} + C + D)$$

The first term must be expanded into $\overline{A} + B + C + D$ and $A + B + C + D$ to get a standard POS expression, which is then mapped; and the cells are grouped as shown in Figure 4–46. The sum term for each group is shown and the resulting minimum POS expression is

$$(C + D)(A + B + D)(\overline{A} + B + C)$$

Keep in mind that this minimum POS expression is equivalent to the original standard POS expression.

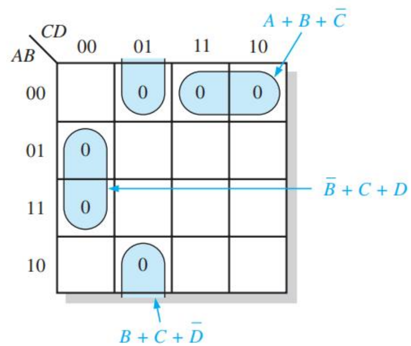


POS and SOP conversion

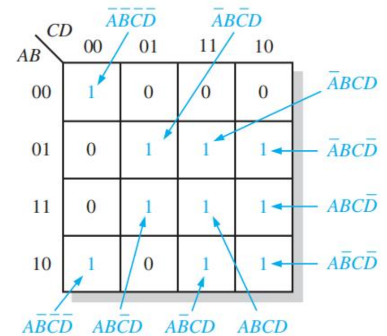
Using a Karnaugh map, convert the following standard POS expression into a minimum POS expression, a standard SOP expression, and a minimum SOP expression.

$$(\overline{A} + \overline{B} + C + D)(A + \overline{B} + C + D)(A + B + C + \overline{D})(A + B + \overline{C} + \overline{D})(\overline{A} + B + C + \overline{D})(A + B + \overline{C} + D)$$

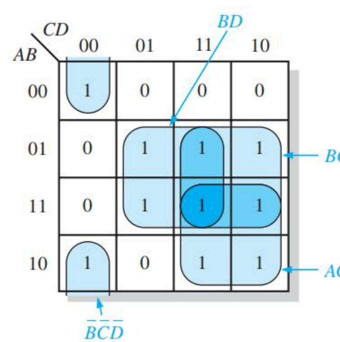
The 0s for the standard POS expression are mapped and grouped to obtain the minimum POS expression in Figure 4–47(a). In Figure 4–47(b), 1s are added to the cells that do not contain 0s. From each cell containing a 1, a standard product term is obtained as indicated. These product terms form the standard SOP expression. In Figure 4–47(c), the 1s are grouped and a minimum SOP expression is obtained.



(a) Minimum POS: $(A + B + \overline{C})(\overline{B} + \overline{C} + D)(B + C + \overline{D})$



(b) Standard SOP:
$\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BCD + \overline{A}BC\overline{D} + ABC\overline{D} + A\overline{B}C\overline{D} + A\overline{B}\overline{C}\overline{D} + AB\overline{C}D + A\overline{B}CD + ABCD$



(c) Minimum SOP: $AC + BC + BD + \overline{B}\overline{C}\overline{D}$

Summary

- SOP simplification: group 1s, write terms in SOP
- POS simplification: group 0s, write terms in POS
- POS and SOP conversion